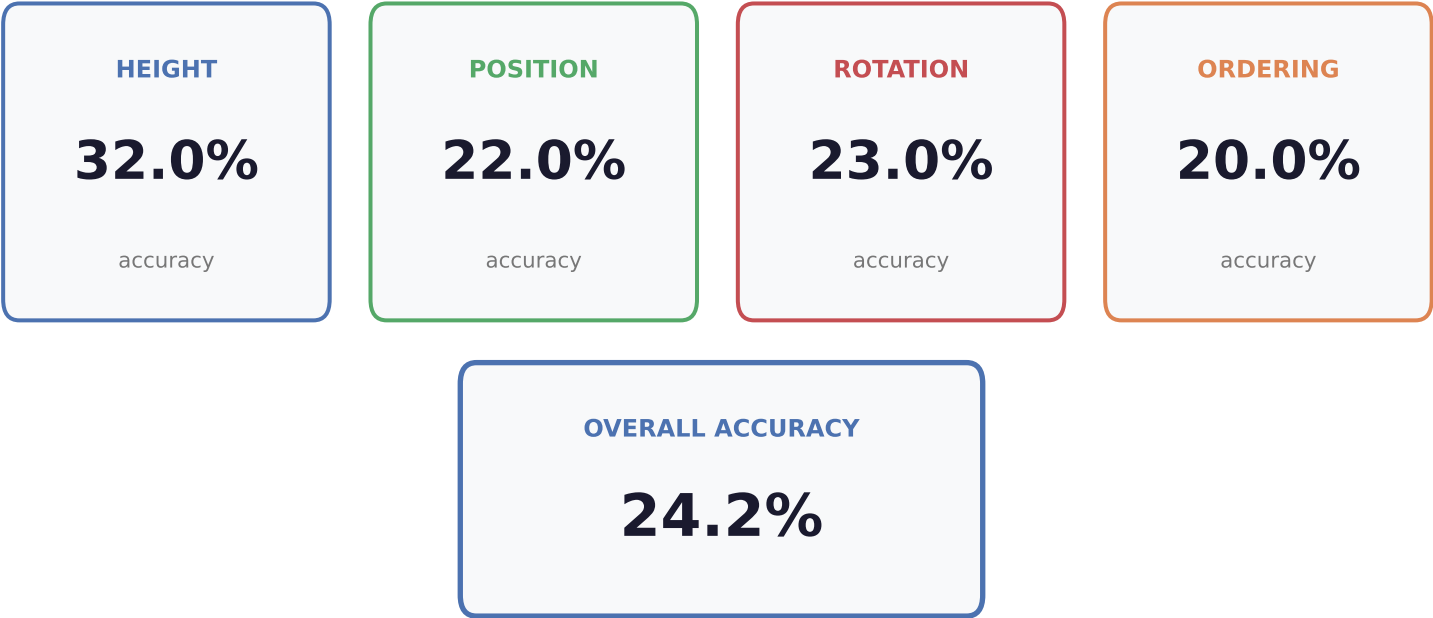


MoE Expert Routing Analysis

Qwen3-VL-30B-A3B-Instruct

LEGOLite (400 questions)



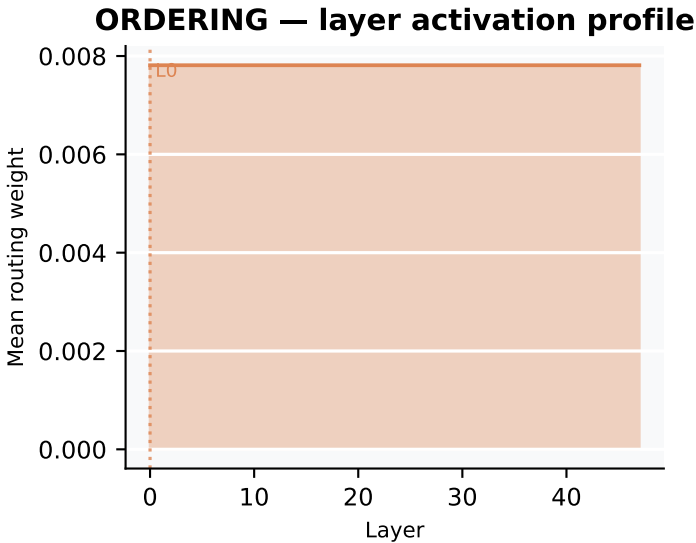
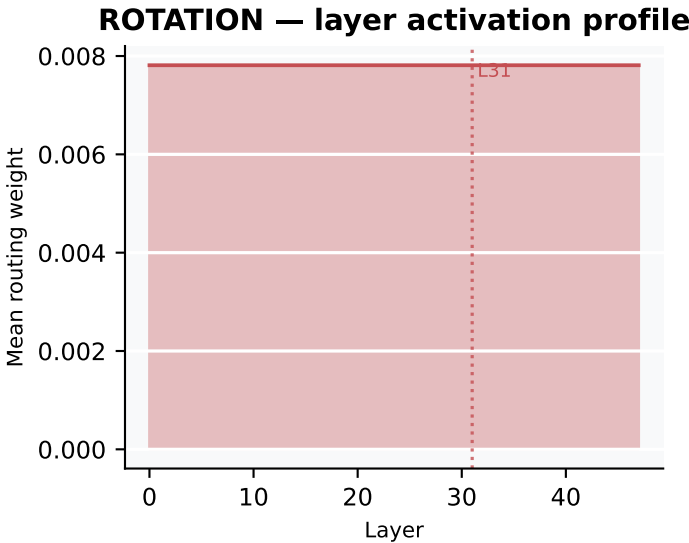
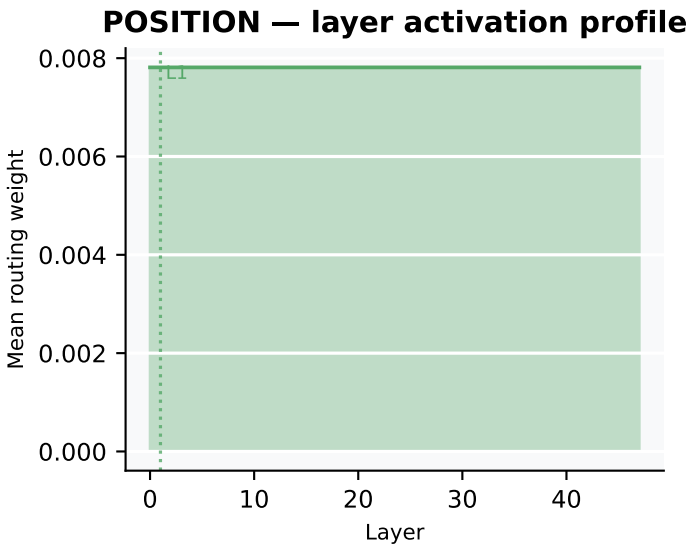
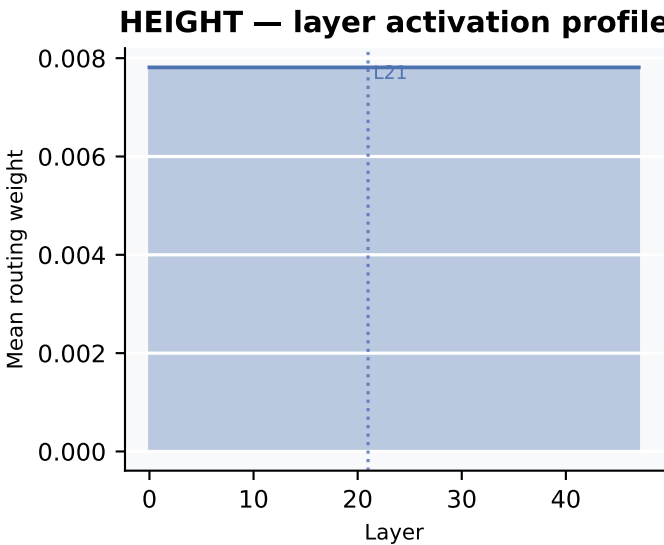
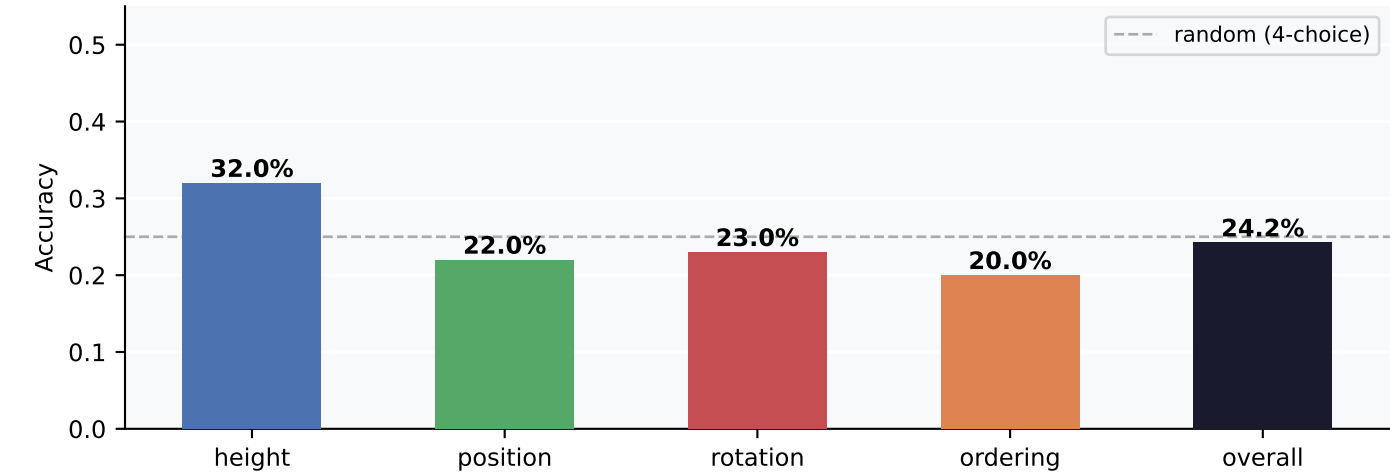
Key Findings

- 48 MoE layers × 128 experts × Top-8 routing analysed
- 400 LEGOLite questions (100 per spatial category)
- Router activations captured for visual tokens only
- Top node: Layer 17 Expert 36 (global $\Delta = +0.0100$)
- Ordering suspects: L4_E13, L2_E37, L10_E3 (negative Δ)
- Height vs Rotation show distinct mid-layer expert clusters

Accuracy & Per-Category Expert Activation

How often the model answers correctly, and which layers are most active per category

Accuracy by Category



Top-30 Spatial Reasoning Experts — Global Leaderboard

Ranked by success_delta = mean_activation_correct – mean_activation_incorrect

Rank	Expert	Act (correct)	Act (incorrect)	Success Δ
1	L17_E36	0.13416	0.12420	+0.00996
2	L8_E21	0.06730	0.05836	+0.00894
3	L12_E94	0.06592	0.05726	+0.00866
4	L10_E65	0.08318	0.07515	+0.00803
5	L19_E64	0.04560	0.03785	+0.00774
6	L18_E7	0.03048	0.02398	+0.00650
7	L7_E107	0.05088	0.04439	+0.00648
8	L4_E101	0.05483	0.04838	+0.00645
9	L7_E64	0.03184	0.02548	+0.00636
10	L14_E41	0.07724	0.07111	+0.00613
11	L14_E51	0.11599	0.11001	+0.00598
12	L24_E121	0.07004	0.06432	+0.00572
13	L5_E44	0.09684	0.09119	+0.00565
14	L19_E83	0.12531	0.11968	+0.00564
15	L20_E23	0.06995	0.06435	+0.00560
16	L22_E65	0.05686	0.05130	+0.00556
17	L19_E41	0.03598	0.03046	+0.00552
18	L40_E126	0.05992	0.05443	+0.00549
19	L45_E45	0.05562	0.05014	+0.00548
20	L32_E21	0.08745	0.08197	+0.00548
21	L27_E126	0.05254	0.04715	+0.00540
22	L17_E62	0.08677	0.08143	+0.00534
23	L16_E44	0.07132	0.06604	+0.00528
24	L28_E101	0.10449	0.09923	+0.00526
25	L7_E41	0.02754	0.02238	+0.00517
26	L16_E37	0.05925	0.05409	+0.00516
27	L12_E121	0.05834	0.05318	+0.00515
28	L20_E21	0.10019	0.09522	+0.00497
29	L12_E44	0.02175	0.01692	+0.00484
30	L26_E51	0.09646	0.09165	+0.00481

Category Specialist Experts

Top-10 experts per spatial category, ranked by category-specific success Δ

HEIGHT specialists

Expert	Act (correct)	Δ height
L16_E37	0.05925	+0.01421
L18_E7	0.03048	+0.01353
L12_E44	0.02175	+0.01243
L7_E64	0.03184	+0.01224
L19_E64	0.04560	+0.01215
L28_E126	0.06153	+0.01103
L40_E126	0.05992	+0.01093
L8_E21	0.06730	+0.01024
L9_E27	0.05769	+0.01018
L9_E5	0.05424	+0.00977

POSITION specialists

Expert	Act (correct)	Δ position
L34_E69	0.00999	+0.01281
L24_E22	0.00980	+0.01223
L37_E45	0.00927	+0.01161
L35_E59	0.00896	+0.01107
L23_E59	0.00900	+0.01095
L25_E45	0.01284	+0.01089
L13_E105	0.07324	+0.00991
L33_E33	0.00791	+0.00977
L15_E58	0.05288	+0.00965
L5_E53	0.06474	+0.00923

ROTATION specialists

Expert	Act (correct)	Δ rotation
L12_E57	0.04335	+0.01224
L5_E44	0.09684	+0.01105
L24_E57	0.02993	+0.00821
L18_E101	0.02959	+0.00762
L10_E65	0.08318	+0.00706
L30_E101	0.02826	+0.00704
L15_E57	0.07742	+0.00650
L8_E43	0.04192	+0.00620
L3_E3	0.01682	+0.00612
L36_E57	0.03965	+0.00597

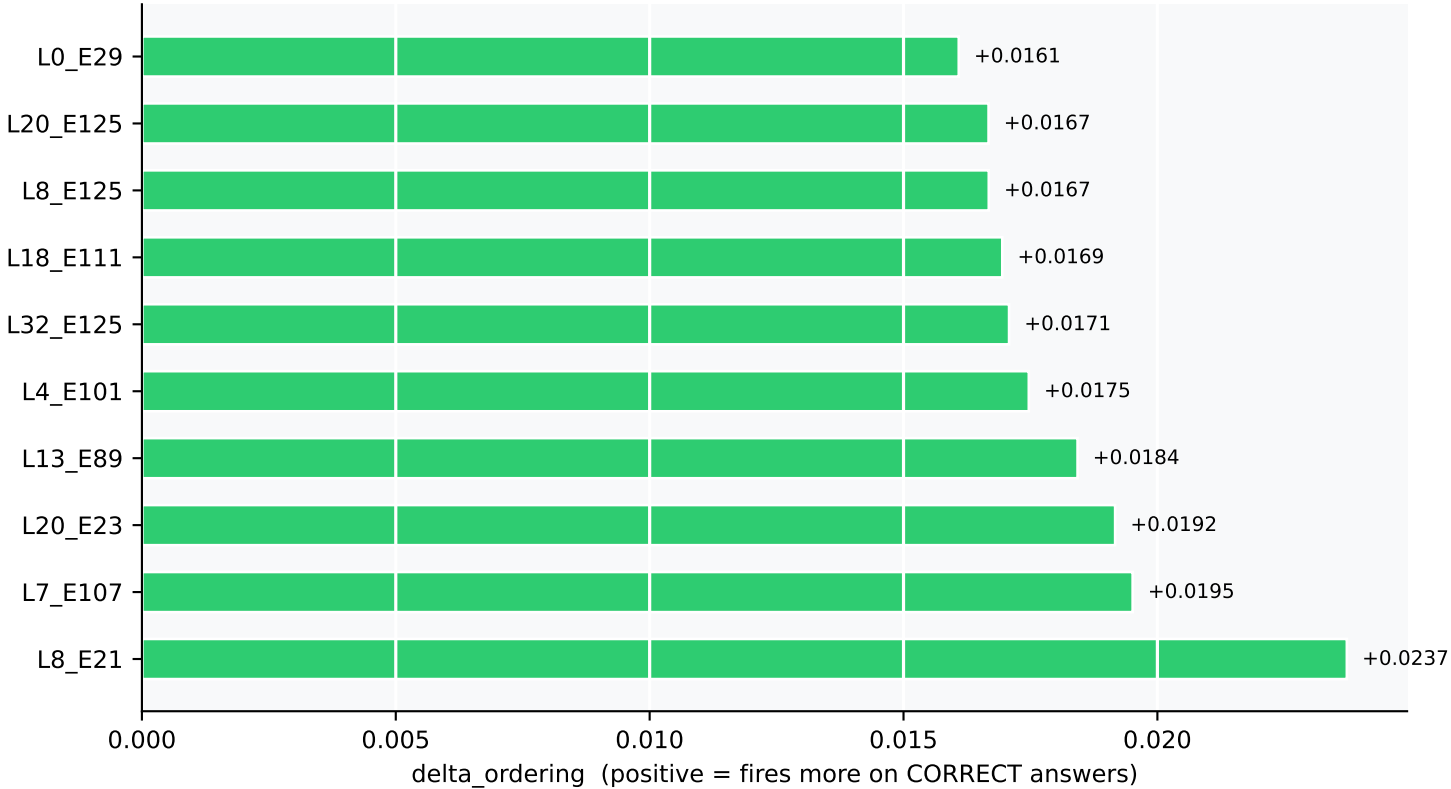
ORDERING specialists

Expert	Act (correct)	Δ ordering
L8_E21	0.06730	+0.02373
L7_E107	0.05088	+0.01951
L20_E23	0.06995	+0.01917
L13_E89	0.06753	+0.01843
L4_E101	0.05483	+0.01747
L32_E125	0.04347	+0.01708
L18_E111	0.06481	+0.01695
L8_E125	0.04741	+0.01668
L20_E125	0.04348	+0.01668
L0_E29	0.03430	+0.01609

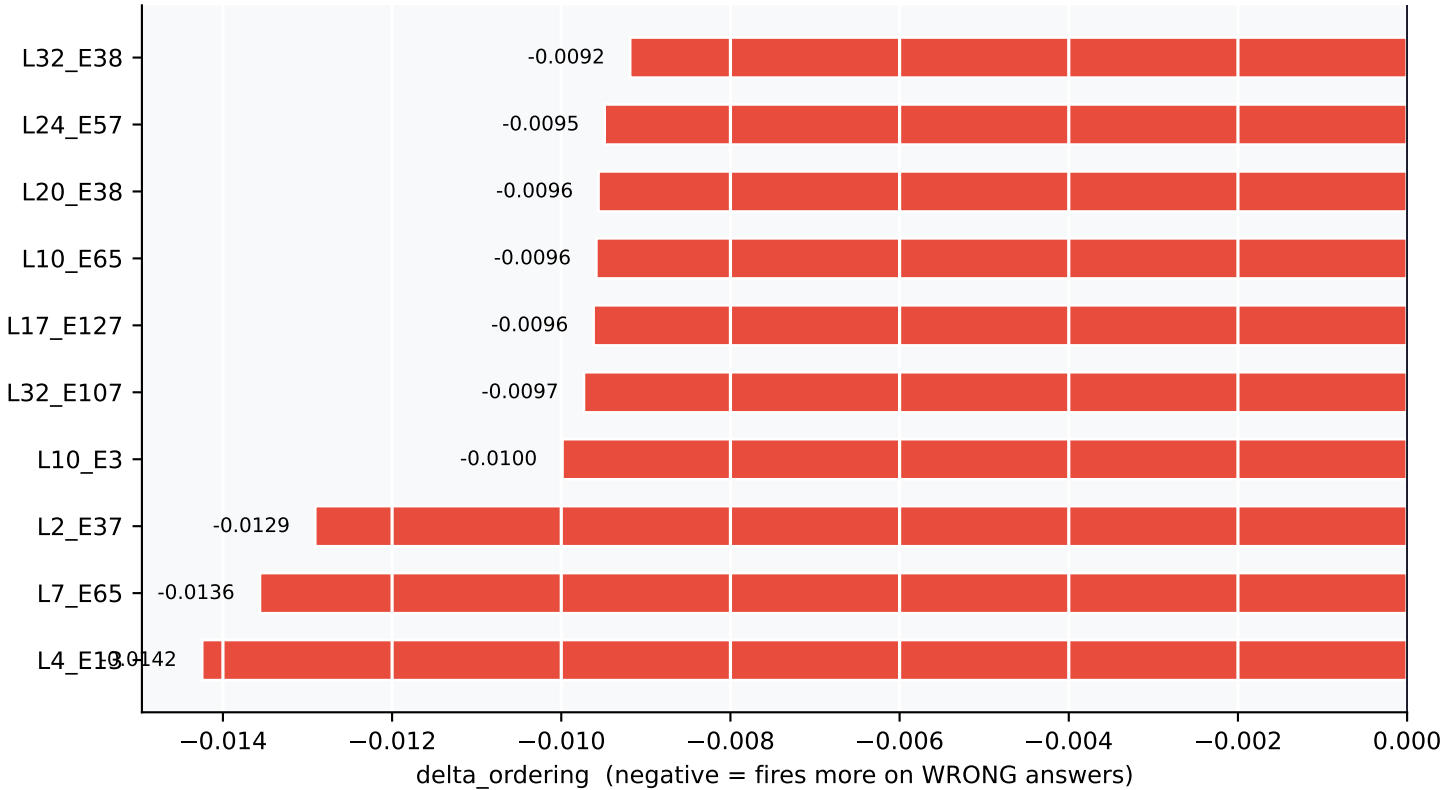
Ordering Task — Expert Failure Diagnosis

Experts that activate heavily on ordering tasks but correlate with WRONG answers

Top Experts Correlated With Correct Ordering Answers



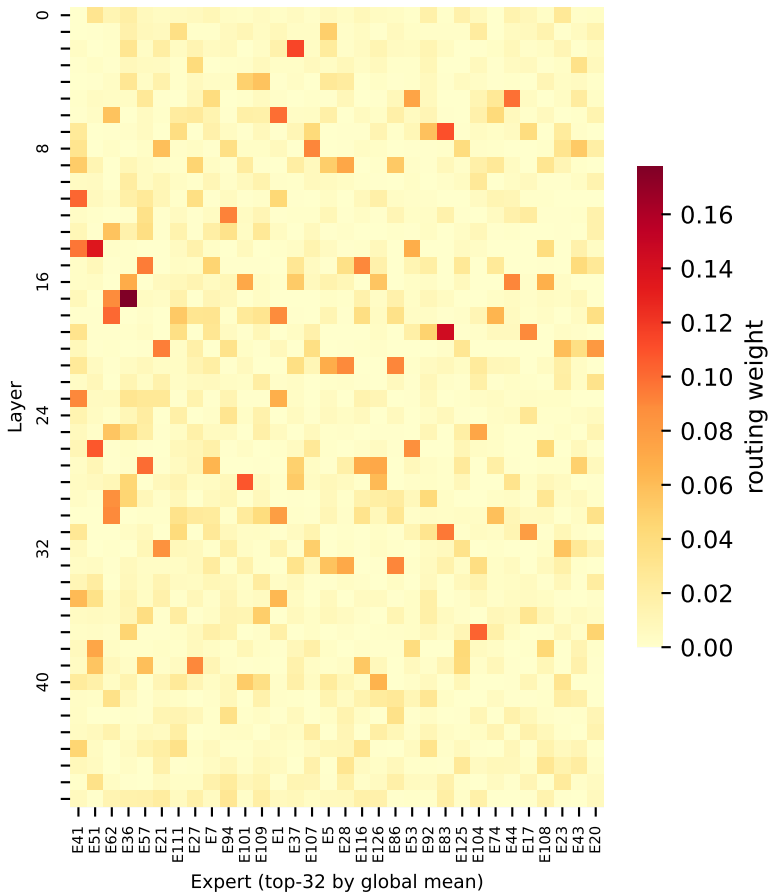
Suspect Experts — Active on Ordering Failures



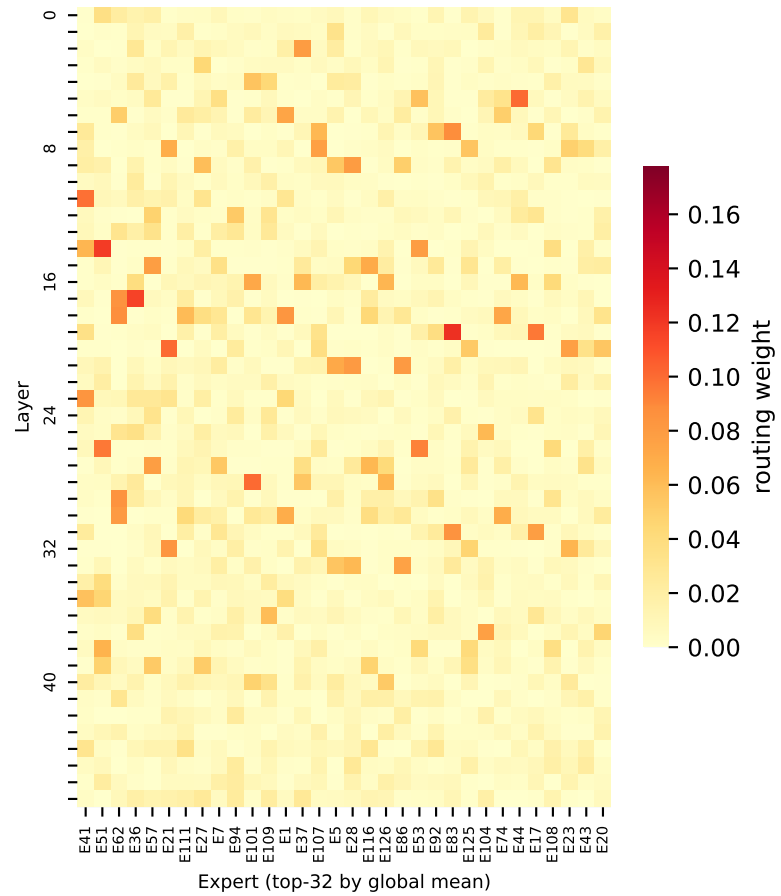
Visual Token Expert Activation Heatmaps

(Layer × Top-32 most active experts; colour = mean routing weight)

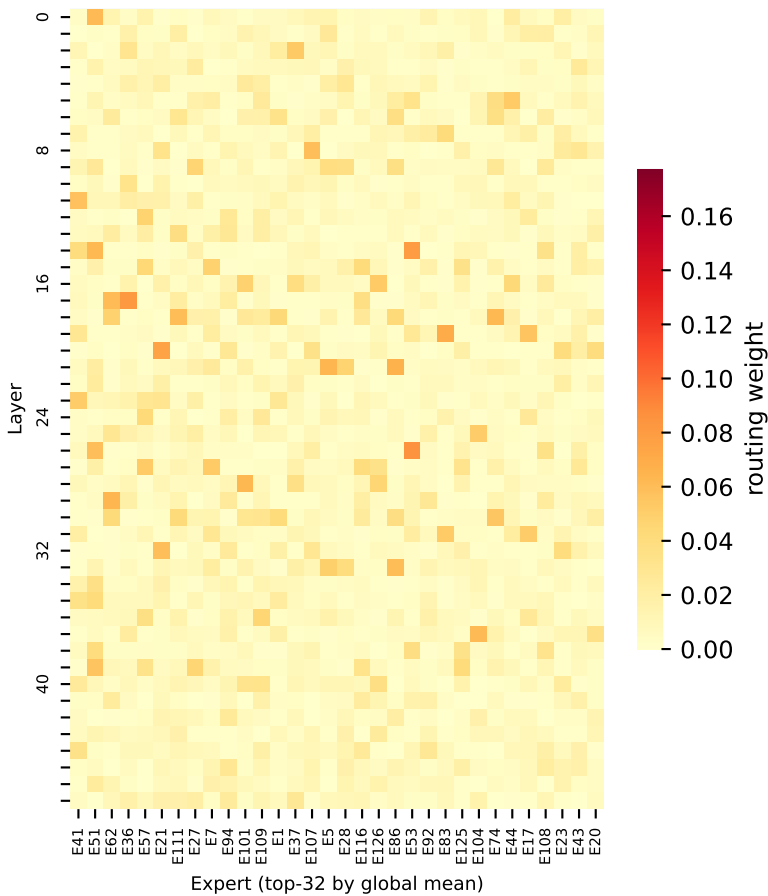
HEIGHT (acc=32.0%)



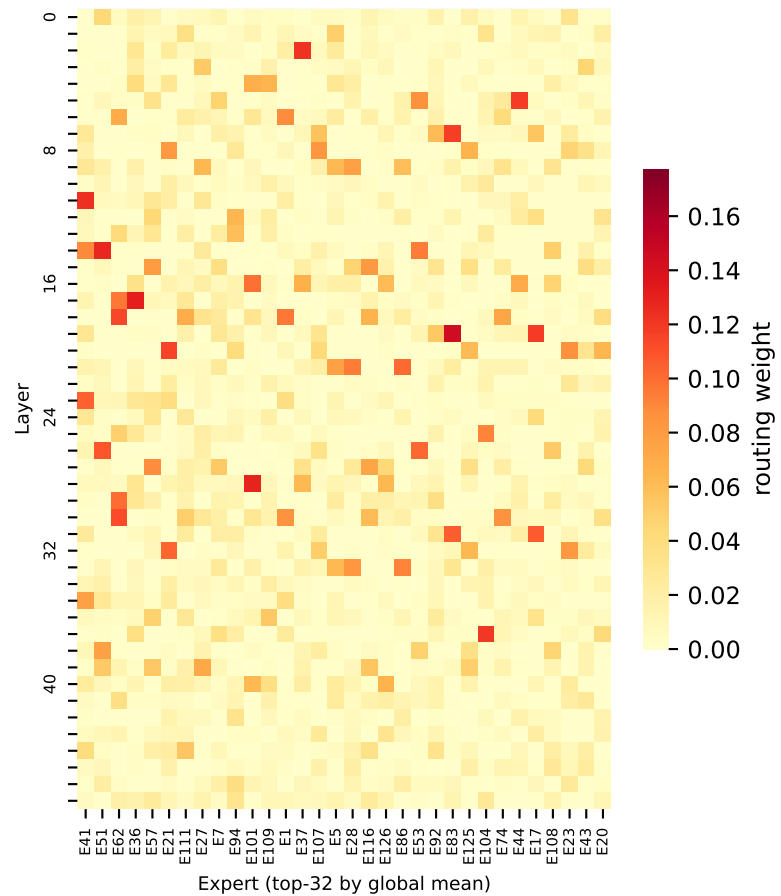
POSITION (acc=22.0%)



ROTATION (acc=23.0%)



ORDERING (acc=20.0%)

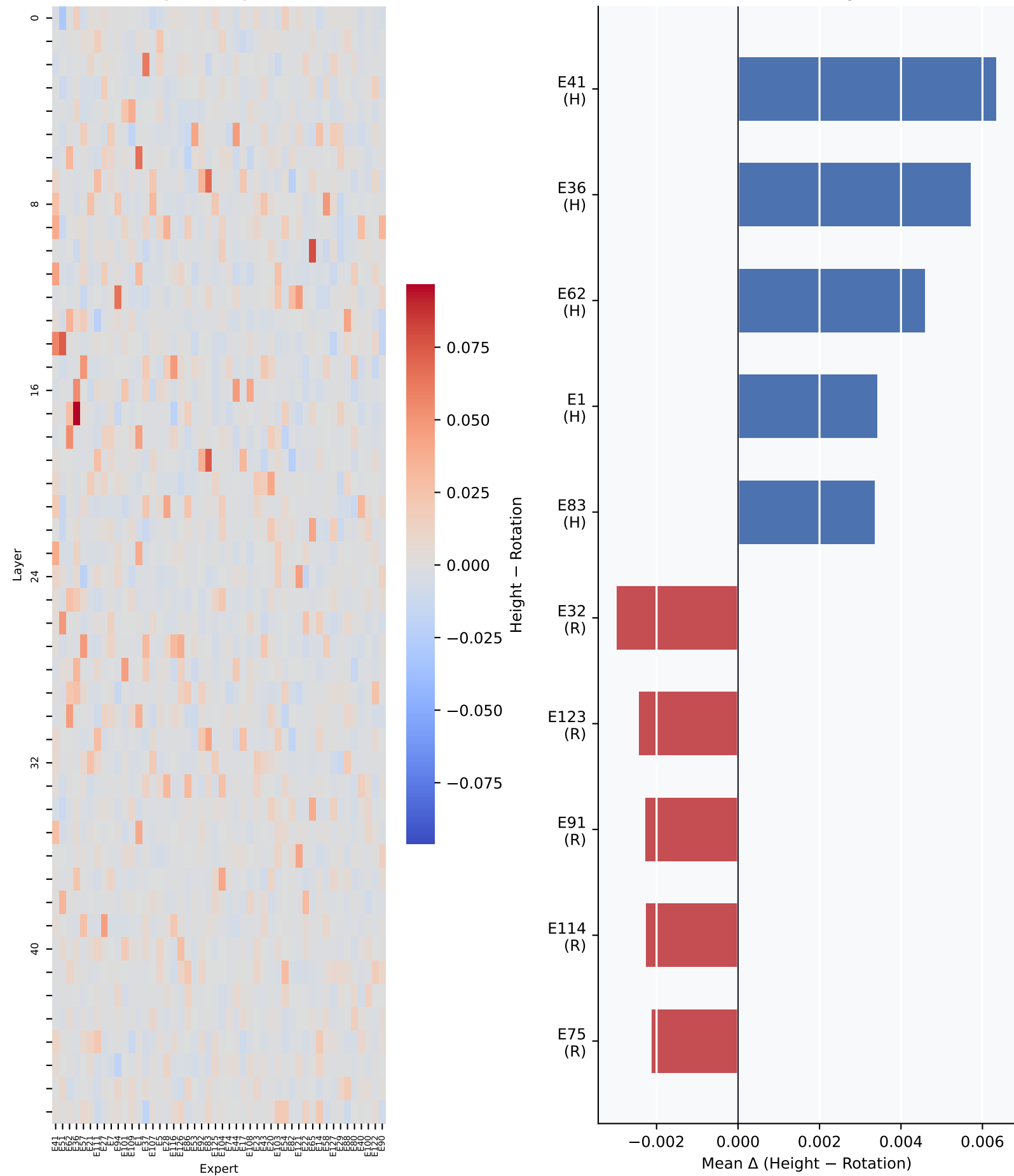


Height vs Rotation — Expert Specialisation

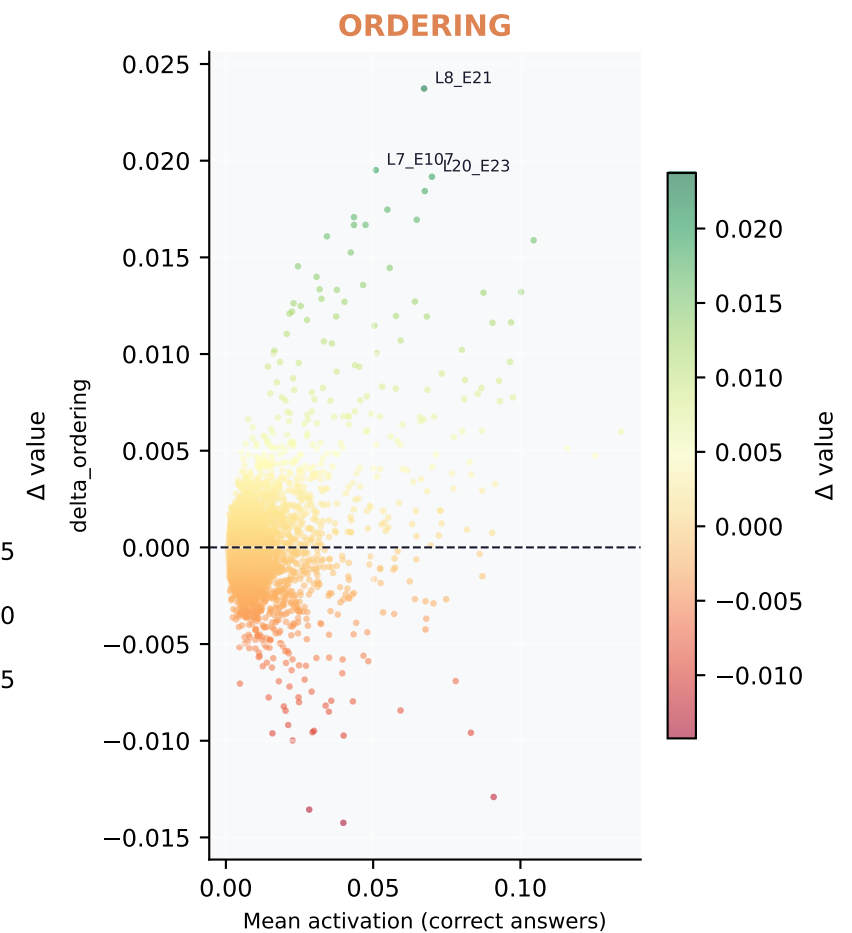
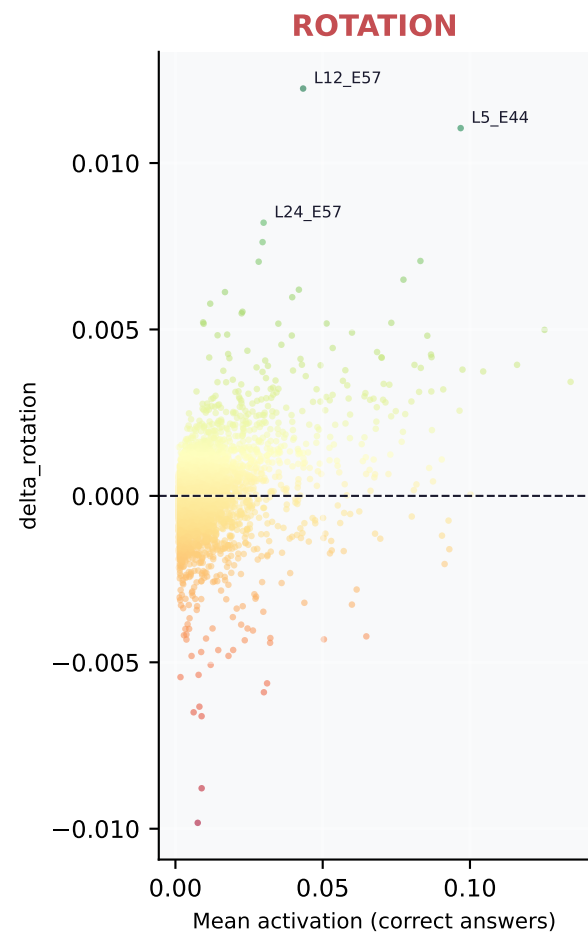
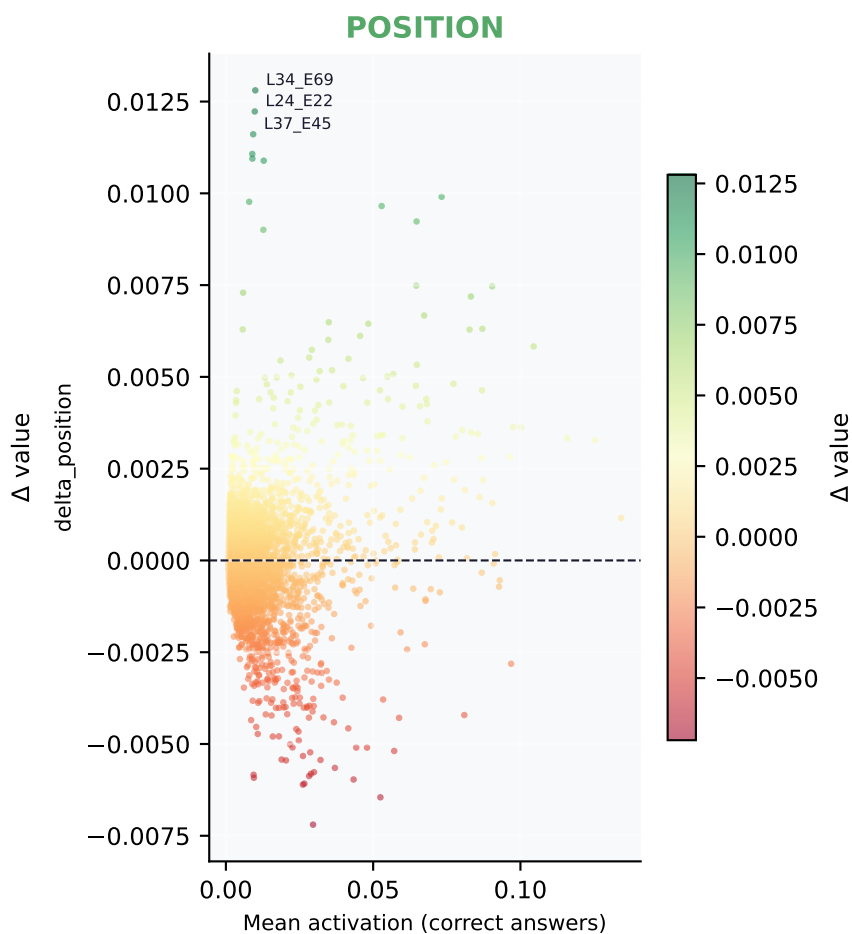
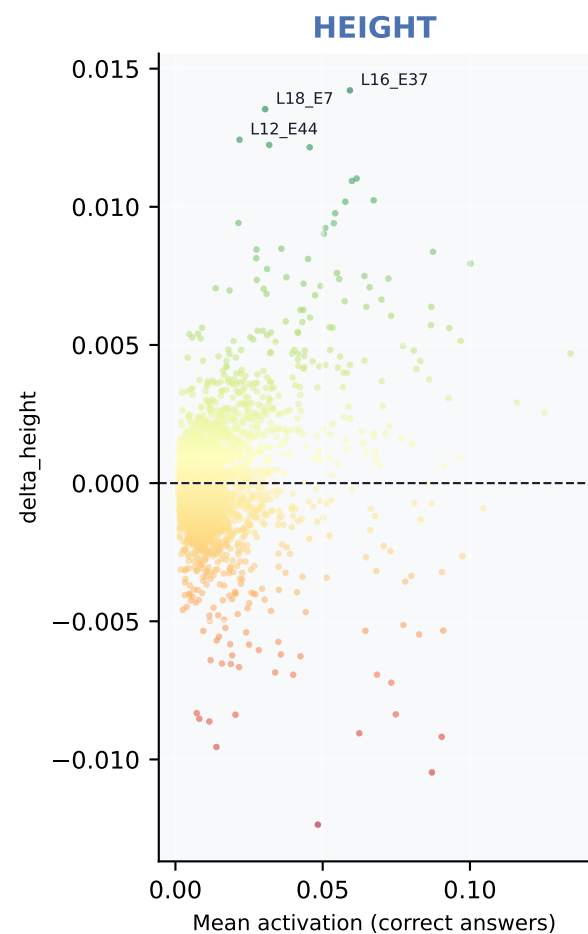
Blue = Height-preferring experts | Red = Rotation-preferring experts | White = shared

**Height – Rotation difference
(top-48 experts)**

**Most specialised experts
across all layers**



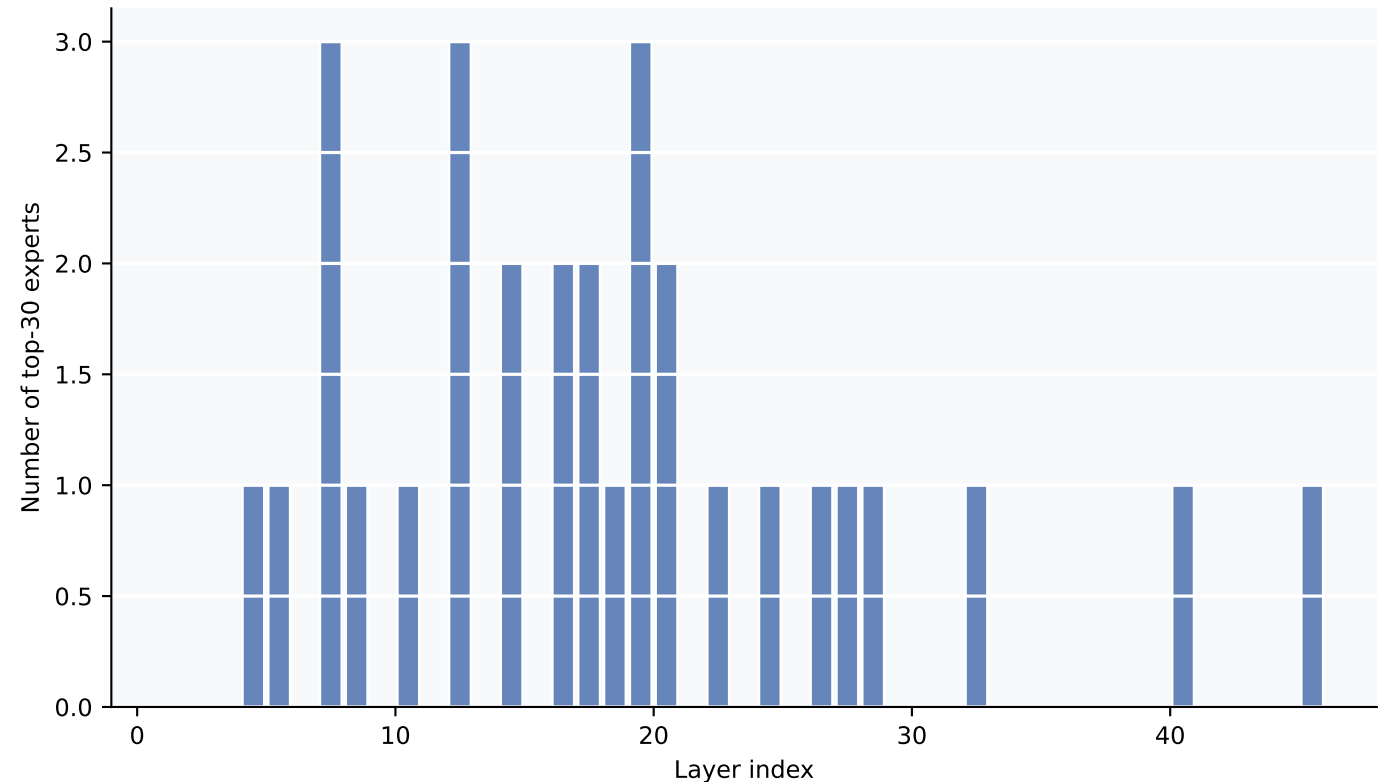
Expert Activation vs Success Δ — per Category
(Top-right quadrant = high activation + positive correlation with correct answers)



Where in the Network Do Spatial Experts Live?

Layer distribution of the top-30 global spatial experts vs category specialists

Layer Distribution — Top-30 Global Spatial Experts



Layer Distribution — Per-Category Top-10 Specialist Experts

